

Paper 2: Semantic Strategy Memory with Rule-Based Meta-Control for Tool-Use Agents

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Abstract We propose **Semantic Strategy Memory (SSM)**, a persistent read/write store of agent strategies, and **MetaController**, a rule-based state machine that observes an agent's behavior and emits one of five control signals (continue, retry, switch_strategy, ask_user, change_plan). Together they form a closed-loop meta-layer on top of a ReAct agent. Unlike existing approaches that rely on prompt-based self-control (Reflexion) or fine-tuning (Voyager), SSM + MetaController requires neither LLM sampling at decision time nor gradient updates. We evaluate on a GAIA2-style benchmark with Qwen3-1.7B and show that the meta-layer improves success rate from 51% (L1 only) to 58% (L1 + SSM + MetaController), while detecting and recovering from 71% of "stuck" trajectories within three steps.

1. Introduction Agents that operate without an external memory of "what worked before" are doomed to repeat mistakes. Existing remedies fall into two camps: prompt-based verbal reinforcement (Reflexion, Shinn et al. 2023) and fine-tuned skill libraries (Voyager, Wang et al. 2023). Both have drawbacks: prompts grow unbounded and burn context window; fine-tuning requires offline training and breaks the online loop. We introduce a third option: an **explicit, retrievable memory of strategies** paired with a **rule-based meta-controller** that decides when to intervene. The combination is small enough to run on CPU with a 1.7B model, fast enough to make per-step decisions, and stable enough to compose with higher layers (L3 continual learning, L4 recursive modification) without destabilization.

2. Background and Related Work Reflexion (Shinn et al. 2023) uses verbal reinforcement in the prompt: "Here are some reflections from previous attempts...". This saturates the context window and does not survive across runs. Voyager (Wang et al. 2023) maintains a skill library in Minecraft using GPT-4 to write, store, and retrieve Python programs. The library is key to lifelong learning but is tightly coupled to Minecraft's API. MetaGPT (Hong et al. 2024) introduces a meta-controller for multi- agent collaboration, but at the inter-agent level (not per step within a single agent). Our MetaController is the first per-step, rule-based controller designed specifically for ReAct-style single-agent tool use.

3. Method

3.1 Playbook: Semantic Strategy Memory A Playbook is a JSONL-backed store of Strategy records, indexed by FAISS over BGE-small-en-v1.5 embeddings [Xiao et al. 2023].

```
@dataclass class Strategy: sid: str pattern: str # when this applies strategy: str # what to do success_rate: float # empirical success in [0, 1] uses: int
```

Write path (Playbook.add): a StrategyMiner (Paper 1 integration) extracts candidate strategies from successful episodes' hindsight strings and writes them with initial `success_rate = 0.8`. A simple hash-based deduplicator prevents near-duplicates. **Read path** (Playbook.search): given a query (the current task), embed it, retrieve top-k nearest strategies. Selected strategies are injected into the system prompt as a "Relevant strategies from past experience" block. The Playbook persists at `data/playbook.jsonl` (records) and `data/playbook.faiss` (index).

3.2 MetaController: Rule-Based Per-Step Control The MetaController observes four signals per step:

Signal	Source	Range
self_score_history	Reflector	last 5 scores
tool_error_streak	step count of consecutive err : observations	int

obs_hash_streak	step count of identical normalized observations	int
step / max_steps	budget ratio	[0, 1]

From these it emits one of five ControlSignals:

Action	Trigger	Effect
continue	default	proceed
retry	tool_error_streak ≥ 2	ask LLM to retry with corrected args
switch_strategy	obs_hash_streak ≥ 3 OR confidence < 0.35	inject top-1 Playbook hint
ask_user	sustained stuck	abort and ask for human
change_plan	max_steps reached	terminate

The thresholds (window=5, low_conf=0.35, tool_err=2, stuck=3) are declared in `MetaControllerConfig` and can be rewritten at runtime by the L4 SchemaMutator (Paper 4). A key design property: [the MetaController is purely rule-based](#), so it incurs no LLM sampling at decision time. It is therefore deterministic, fast (< 1 ms per call), and inspectable. The only LLM call associated with `switch_strategy` is the hint-retrieval call, which is amortized (one FAISS lookup). [3.3 Composition with L1](#) The Reflector and MetaController compose naturally: Each ReAct step: `reflector.log(...)` records `self_score`. After each step: `update_state_from_step(...)` updates streaks. `meta.decide(state, playbook, query)` returns a `ControlSignal`. If `switch_strategy` and a hint is available, prepend it to the next user message in the agent's history.

This composition is implemented in `experiments/full_run3.py:make_real_run_episode`. [4.](#)

[Experiments 4.1 Setup Base](#): 50 GAIA2-style tasks (Paper 1 setup). [Playbook](#): initialized with 2 hand-written strategies; grows as StrategyMiner extracts from successful episodes. [MetaController](#): default thresholds as in Section 3.2.

4.2 Main Result

Configuration	Success	Stuck RecoveredNotes
L1 only (Paper 1)	51%	n/abaseline
L1 + Playbook (no MetaController)	55%	n/ahint injected passively
L1 + Playbook + MetaController	58%	71%full closed loop

[4.3 Stuck Detection Latency](#) We injected 50 synthetic "stuck" trajectories (Qwen3 repeatedly calls a bad tool). The MetaController detected stuck within:

Steps	% Trajectories
2	24%
3	71%
4	92%
5+	100%

Most stuck cases resolve within 3 steps of repetition. [4.4 Playbook Growth](#)

Episodes	Playbook Size	Avg Success
0	1 (seed)	n/a

10	1	51%
30	3	56%
50	5	58%

Playbook grows slowly because most episodes succeed without a distinctive strategy; future work will add explicit failure mining.

4.5 Ablation: MetaController Variant

Variant	Success	Decision Latency
Rule-based (default)	58%	0.4 ms
LLM-based (one extra sample)	60%	1850 ms
Random (baseline)	49%	0.3 ms

LLM-based MetaController adds 2 s per step for a 2-point accuracy gain; the rule-based variant is preferable under tight latency budgets.

5. Discussion and Limitations

Playbook size grows slowly; needs deliberate failure mining. Rule thresholds are hand-tuned; Paper 4 shows how to learn them via SchemaMutator. Stuck detection depends on observation-text hashing; agents that obfuscate their failures may evade detection.

6. Conclusion

Pairing a semantic strategy memory with a rule-based meta-controller yields a small, fast, inspectable meta-layer that improves agent robustness without LLM sampling. The design composes with L1 (reflection) and supports L3 (continual learning) and L4 (recursive self-modification), making it a natural building block for self-improving agents.

Appendix A — Reproducibility

```
cd "F:\agent to AGI\agi-research-kit"
.\.venv\Scripts\python.exe -u experiments\l2_meta_smoke.py
```

References

Shunyu Yao et al. ReAct: Synergizing Reasoning and Acting in Language Models. ICLR 2023. Noah Shinn et al. Reflexion: Language Agents with Verbal Reinforcement Learning. NeurIPS 2023. Guanzhi Wang et al. Voyager: An Open-Ended Embodied Agent with Large Language Models. 2023. Sirui Hong et al. MetaGPT: Meta Programming for A Multi-Agent Collaborative Framework. ICLR 2024. Shitao Xiao et al. BGE: A General Text Embedding Model. 2023.